

$$f(t) = e^{i\omega_0 t}$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt e^{i\omega_0 t} e^{-i\omega t} = \sqrt{2\pi} \delta(\omega - \omega_0)$$

Reverse:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \sqrt{2\pi} \delta(\omega - \omega_0) e^{+i\omega t} = e^{i\omega_0 t} \quad \checkmark$$

$$\mathcal{F}(e^{i\omega_0 t}) = \sqrt{2\pi} \delta(\omega - \omega_0)$$

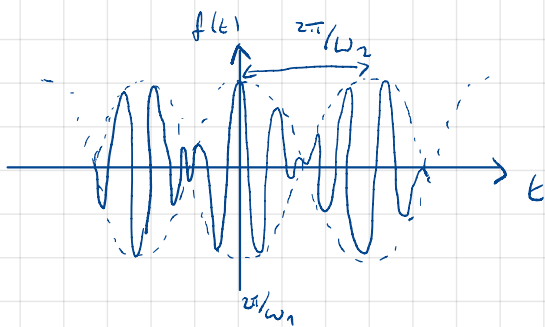
$$f(t) = \cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t})$$

$$\mathcal{F}(\cos \omega_0 t) = \sqrt{\frac{\pi}{2}} (\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$$

Amplitude modulation (AM):

$$f(t) = \cos(\omega_1 t) \cos(\omega_2 t) = g_1(t) g_2(t)$$

$\omega_1 \gg \omega_2$:



$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \hat{g}_1(\omega) * \hat{g}_2(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega' \hat{g}_1(\omega - \omega') \hat{g}_2(\omega')$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{\pi}{2} \int_{-\infty}^{\infty} d\omega' \left[\delta(\omega - \omega' - \omega_1) + \delta(\omega - \omega' + \omega_1) \right] \left[\delta(\omega' - \omega_2) + \delta(\omega' + \omega_2) \right]$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{\pi}{2} \int_{-\infty}^{\infty} d\omega' \left[\delta(\omega - \omega' - \omega_1) \delta(\omega' - \omega_2) + \delta(\omega - \omega' - \omega_1) \delta(\omega' + \omega_2) \right. \\ \left. + \delta(\omega - \omega' + \omega_1) \delta(\omega' - \omega_2) + \delta(\omega - \omega' + \omega_1) \delta(\omega' + \omega_2) \right]$$

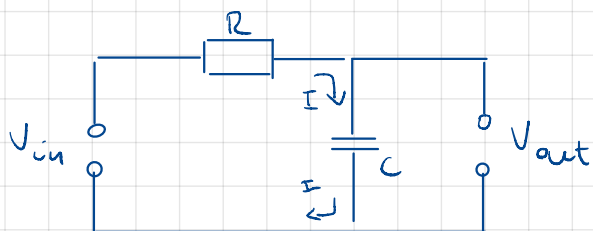
$$= \frac{1}{\sqrt{2\pi}} \pi/2 \left[\delta(\omega - \omega_2 - \omega_1) + \delta(\omega + \omega_2 - \omega_1) \right. \\ \left. + \delta(\omega - \omega_2 + \omega_1) + \delta(\omega + \omega_2 + \omega_1) \right]$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}} \sqrt{\frac{\pi}{2}}}_{1/2} \mathcal{F}[\cos(\omega_1 + \omega_2)t + \cos(\omega_1 - \omega_2)t]$$

$$\Rightarrow \cos(\omega_1 t) \cos(\omega_2 t) = \frac{1}{2} [\cos(\omega_1 + \omega_2)t + \cos(\omega_1 - \omega_2)t] \quad \checkmark$$



low pass filter:



$$\frac{Q}{C} + RI = V_{in}$$

$$\frac{Q}{C} = V_{out}$$

$$\frac{Q}{C} + R \cdot \dot{Q} = V_{in}$$

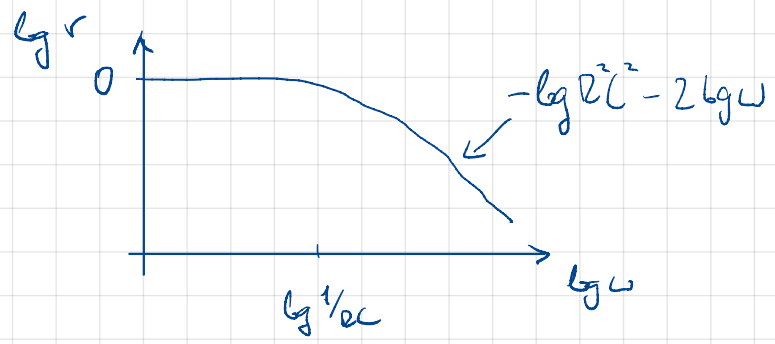
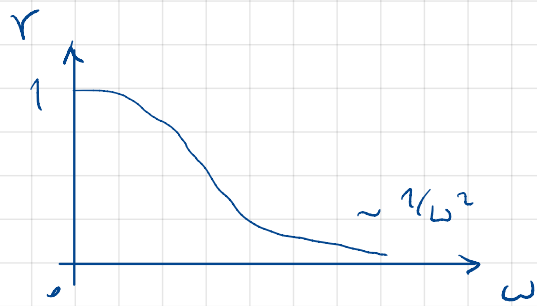
$$\frac{1}{C} \hat{Q}(\omega) + R \cdot i\omega \hat{Q}(\omega) = \hat{V}_{in}(\omega)$$

$$\hat{Q}(\omega) = \frac{\hat{V}_{in}(\omega)}{\frac{1}{C} + i\omega R}$$

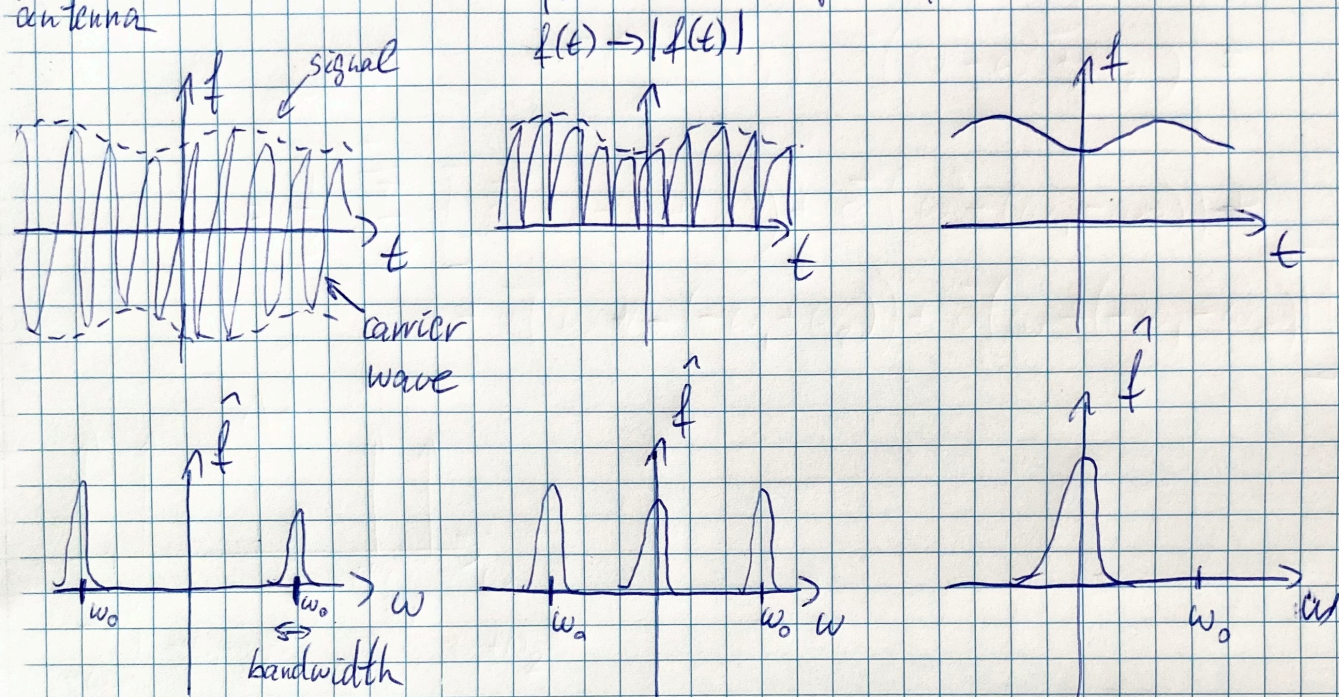
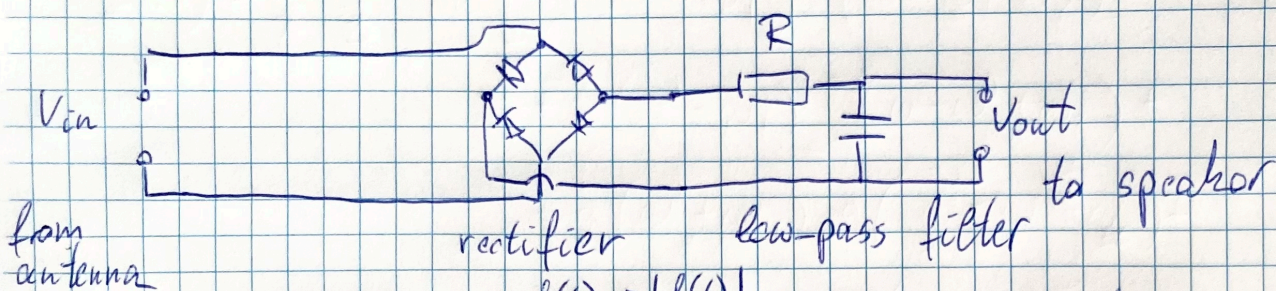
$$\hat{V}_{out} = \frac{1}{C} \hat{Q}(\omega) = \frac{\hat{V}_{in}(\omega)}{1 + i\omega RC}$$

$$\gamma = \frac{P_{out}}{P_{in}} = \frac{|\hat{V}_{out}|^2}{|\hat{V}_{in}|^2} = \frac{1}{1 + \omega^2 R^2 C^2}$$

$$\begin{aligned} \omega \ll \frac{1}{RC} &\rightarrow \approx 1 - \omega^2 R^2 C^2 \\ \omega \gg \frac{1}{RC} &\rightarrow \approx \frac{1}{R^2 C^2 \omega^2} \end{aligned}$$



AM radio receiver



Low pass filter

$$\hat{V}_{out}(\omega) = \hat{V}_{in}(\omega) \frac{1}{1 + i\omega RC}$$

$$f_{LPF}(t) = \begin{cases} e^{-\frac{t}{RC}} & \text{for } t > 0 \\ 0 & t < 0 \end{cases}$$

response function

$$\hat{V}_{out}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt' V_{in}(t-t') f_{LPF}(t')$$

Convolution is generalized moving average,
smoothens out fast variations

Some Fourier transforms

1.)

$$f(x) = \begin{cases} 1 & -b < x < c \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_b^c dx e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-i\omega} e^{-i\omega x} \right]_{x=b}^c = \\ &= \frac{i}{\sqrt{2\pi}\omega} (e^{-i\omega c} - e^{-i\omega b}) \end{aligned}$$

If $b = -c$ then $\hat{f}(\omega) = \frac{i}{\sqrt{2\pi}\omega} (e^{-i\omega c} - e^{i\omega c}) = \sqrt{\frac{2}{\pi}} \frac{\sin(\omega c)}{\omega}$

2.)

$$f(x) = \begin{cases} e^{-ax} & (a > 0) \quad x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (*)$$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x} \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(a+i\omega)x}}{-(a+i\omega)} \right]_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}} \left(0 - \frac{1}{-(a+i\omega)} \right) = \frac{1}{\sqrt{2\pi}} \frac{1}{a+i\omega} \end{aligned}$$

3.)

$$g(x) = e^{-a|x|} \quad a > 0$$

Note: $g(x) = f(x) + f(-x)$ where $f(x)$ is given in (*)

$$\begin{aligned} \text{Then } \hat{g}(\omega) &= \hat{f}(\omega) + \hat{f}(-\omega) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a+i\omega} + \frac{1}{a-i\omega} \right) \\ &= \frac{1}{\sqrt{2\pi}} \frac{2a}{a^2 + \omega^2} \end{aligned}$$

or:

$$f(x) = e^{-a|x|} \quad a > 0$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-a|x|} e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 dx e^{+ax} e^{-i\omega x} + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x}$$

$$\begin{aligned} & y = -x \\ & = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dy e^{-ay} e^{i\omega y} + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-ax} e^{-i\omega x} \end{aligned}$$

$$\text{Here } \int_0^{\infty} dx e^{-ax} e^{-i\omega x} = \left[\frac{e^{-ax-i\omega x}}{-a-i\omega} \right]_{x=0}^{\infty} = 0 - \frac{1}{-a-i\omega} = \frac{1}{a+i\omega}$$

1st term: $\omega \rightarrow -\omega$, so

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a-i\omega} + \frac{1}{a+i\omega} \right) = \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2}$$

Reverse direction:

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \hat{f}(\omega) e^{+i\omega x}$$

So,

$$e^{-a|x|} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2} e^{i\omega x} \stackrel{\omega \rightarrow -\omega}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2} e^{-i\omega x}$$

swapping ω and x :

$$\mathcal{F}\left(\frac{1}{x^2 + a^2}\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \frac{e^{-i\omega x}}{x^2 + a^2} = \sqrt{\frac{\pi}{2}} \frac{e^{-a|\omega|}}{a}$$

We don't need to compute this integral!

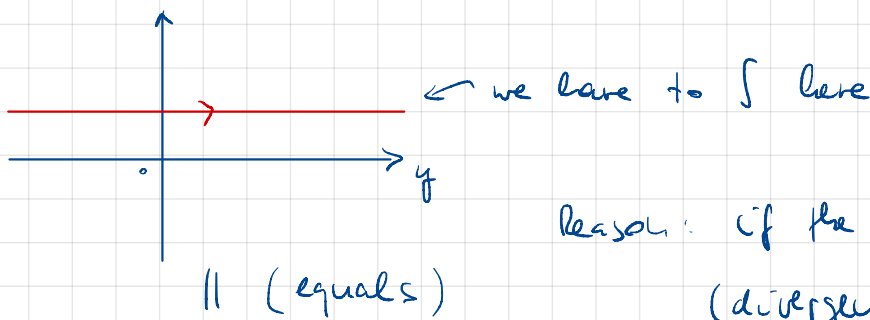
4.) $f(x) = e^{-ax^2}$ ($a > 0$) Gaussian (bell curve)

$$\sqrt{2\pi} \hat{f}(\omega) = \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-i\omega x} = \int_{-\infty}^{\infty} dx e^{-a\left(x + \frac{i\omega}{2a}\right)^2 + \left(\frac{\omega}{2a}\right)^2}$$

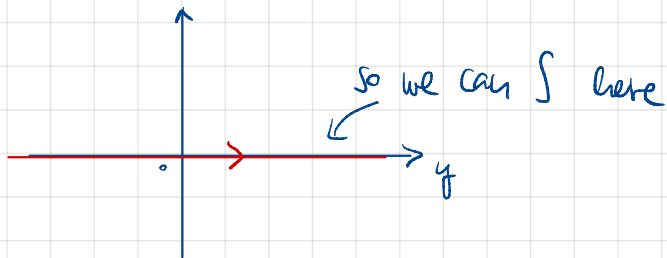
$$= e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} dx e^{-a\left(x + \frac{i\omega}{2a}\right)^2}$$

Naïve change of variables: $y = x + \frac{i\omega}{2a}$

$$= e^{-\frac{\omega^2}{4a}} \int_{-\infty + \frac{i\omega}{2a}}^{\infty + \frac{i\omega}{2a}} dy e^{-ay^2}$$



Reason: if the integrand has no poles (divergences) and no analytic points the integration contour can be shifted



Now we use

$$\int_{-\infty}^{\infty} dy e^{-ay^2} = \sqrt{\frac{\pi}{a}} \quad (\text{useful to know by heart!})$$

Proof: Consider the square:

$$\int_{-\infty}^{\infty} dy e^{-ay^2} \cdot \int_{-\infty}^{\infty} dz e^{-az^2} = \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz e^{-a(y^2+z^2)}$$

Integral on y - z plane, integrand depends on $r^2 = y^2 + z^2$ only.

Use radial coordinates:

$$= 2\pi \int_0^{\infty} dr r e^{-ar^2} = 2\pi \left[-\frac{1}{2a} e^{-ar^2} \right]_{r=0}^{\infty} = 2\pi \left(0 - \left(-\frac{1}{2a}\right) \right) = \frac{\pi}{a}$$

So,

$$F(e^{-ax^2}) = \hat{f}(\omega) = \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} = \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}}$$

5.) $f(x) = e^{-a(x-b)^2}$ $a > 0$

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-a(x-b)^2} e^{-i\omega x} \stackrel{z=x-b}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-az^2} e^{-i\omega(z+b)} = \\ &= \frac{e^{-i\omega b}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-az^2} e^{-i\omega z} = e^{-i\omega b} \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} \end{aligned}$$

[General rule: $g(x) = f(x-b) \Rightarrow \hat{g}(\omega) = e^{-i\omega b} \hat{f}(\omega)$]

Reverse direction:

$$e^{-a(x-b)^2} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} e^{-i\omega b} \cdot e^{+i\omega x} \stackrel{\omega=-\omega}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \frac{e^{-\frac{\omega^2}{4a}}}{\sqrt{2a}} e^{+i\omega b} e^{-i\omega x}$$

$\begin{matrix} \omega \rightarrow x \\ x \rightarrow \omega \end{matrix} \rightarrow F\left(e^{-\frac{x^2}{4a}} e^{+ibx}\right) = \sqrt{2a} e^{-a(\omega-b)^2}$

Now directly:

6.) $h(x) = e^{-ax^2 + ibx}$

$$\hat{h}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-ax^2 + ibx} e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-i(\omega-b)x}$$

from Problem 4

$$= \hat{f}(\omega-b) = \frac{e^{-\frac{(\omega-b)^2}{4a}}}{\sqrt{2a}}$$

With $a = 1/4a$ we recover the result in 5).

$$7) f_n(x) = x^n e^{-ax^2} \quad n \in \mathbb{Z}^+, \quad a > 0$$

A general rule:

$$\begin{aligned} \mathcal{F}[x \cdot g(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \, x g(x) e^{-i\omega x} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx g(x) \cdot i \frac{d}{d\omega} (e^{-i\omega x}) \\ &= i \frac{d}{d\omega} \mathcal{F}[g(x)] = i \frac{d\hat{g}(\omega)}{d\omega} \end{aligned}$$

So,

$$n=1: \quad \hat{f}_1(\omega) = i \frac{d}{d\omega} \left(\frac{e^{-\omega^2/4a}}{\sqrt{2a}} \right) = \frac{i}{\sqrt{2a}} e^{-\omega^2/4a} \cdot \frac{-2\omega}{4a} = -i\omega \frac{e^{-\omega^2/4a}}{(2a)^{3/2}}$$

$$\begin{aligned} n=2: \quad \hat{f}_2(\omega) &= i \frac{d}{d\omega} \hat{f}_1(\omega) = \frac{e^{-\omega^2/4a}}{(2a)^{3/2}} + \frac{\omega}{(2a)^{3/2}} e^{-\omega^2/4a} \cdot \frac{-2\omega}{4a} \\ &= (2a - \omega^2) \frac{e^{-\omega^2/4a}}{(2a)^{5/2}} \end{aligned}$$

$$8.) f_n(x) = x^n \cdot e^{-a|x|} \quad n \in \mathbb{Z}^+, \quad a > 0$$

$$\begin{aligned} \hat{f}_1(\omega) &= i \frac{d}{d\omega} \left(\frac{1}{\sqrt{2\pi}} \frac{2a}{\omega^2 + a^2} \right) = \frac{i}{\sqrt{2\pi}} \cdot 2a \frac{-1}{(\omega^2 + a^2)^2} \cdot 2\omega \\ &= \frac{-1}{\sqrt{2\pi}} \frac{4ia\omega}{(\omega^2 + a^2)^2} \end{aligned}$$

$$\hat{f}_2(\omega) = \dots$$

Note:

$$\mathcal{F}[x \cdot g(x)] = i \frac{d}{d\omega} \hat{g}(\omega)$$

is the pair of

$$\mathcal{F}^{-1}[\hat{g}'(\omega)] = -ix g(x)$$

$$\mathcal{F}[f'(x)] = i\omega \hat{f}(\omega)$$